

EXECUTIVE SUMMARY

OVERALL PERFORMANCE GRADE: F

Critical — Safety and Technique Review Required

SCORE : 39.2%

CONSISTENCY	LINE QUALITY	BRAKING SCORE	EXIT SPEED
0%	100%	32.3%	24.5%

Laps Completed:	1	Best Lap:	1 : 38.542
Consistency:	72%	Average Lap:	2 : 31.849
Top Speed:	162.0 KM/H	Best Lap #:	2

WHAT YOU DID RIGHT [SUCCESS]

+ Your line through complex turns is excellent. You're using full track width and maintaining apex speed.

+ Upshifting and powerband engagement are crisp and consistent with zero over-rev events.

+ Your best lap shows strong pace capability across all sectors.

TOP COACHING PRIORITY [MAX LAP TIME GAIN]

IMPROVE THRESHOLD BRAKING AT TURN 2

Using 6% of available braking capability (0.09G peak).

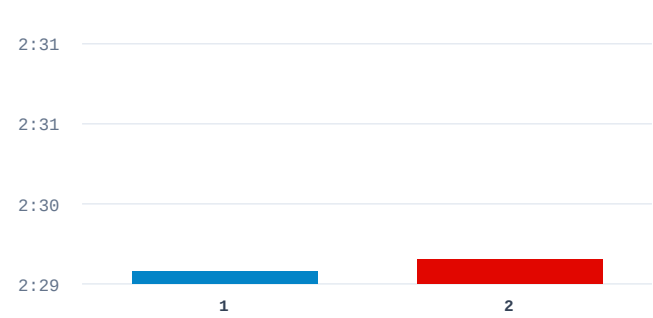
SKIP BARBER COACHING PRINCIPLE

"If you're braking at the 300 mark with no problem, you've got to take small steps to find where that limit is." Danny Sullivan, Going Faster!, Ch.1

ACTION PLAN: Push harder on the brake pedal. Commit to maximum straight-line deceleration before the turn-in, then trail it off progressively.

STINT OVERVIEW - HOW TO READ YOUR LAPS

LAP TIME CONFORMANCE

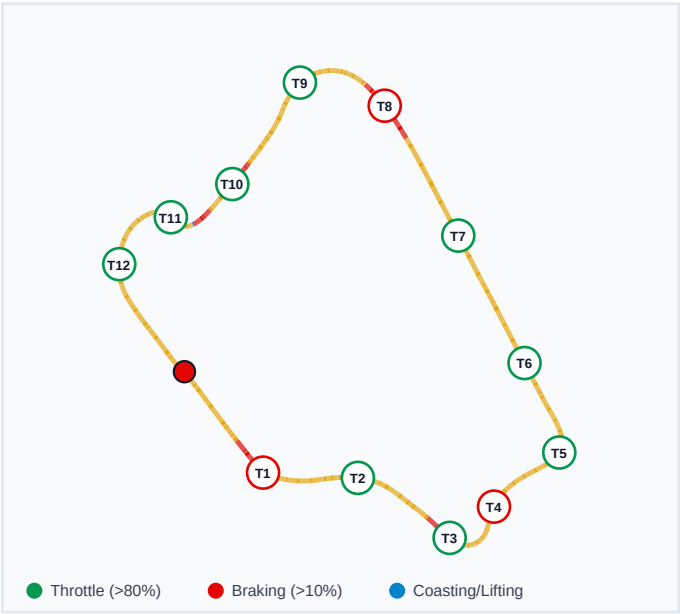


SECTOR LOSS MATRIX

SECTOR	BEST	AVG	LOSS
Sector 1 (T1-T3)	47.8s	48.2s	+0.4s
Sector 2 (T4-T7)	44.2s	45.1s	+0.9s
Sector 3 (T8-T10)	39.5s	40.4s	+0.9s

Note: Sector 2 & 3 hold 1.8s in opportunities.

2D TRACK ANALYSIS MAP



STINT PERFORMANCE SYNTHESIS

The lap comparison chart confirms a structured stint sequence. Sector 2 is your highest priority segment where Turn 7 (Type II corner) causes entry decelerations to lose time. In Sector 3, Turn 9 (Type I carousel corner) suffers from delayed exit acceleration, costing compounding straighta...

LAP-BY-LAP & TURN-BY-TURN TELEMETRY ANALYSIS

"GOING FASTER!" CORNER TAXONOMY & RACECRAFT OBJECTIVES

TYPE I: EXIT SPEED PRIORITY

Leads onto a straight. Exit velocity compounds down straight. Squeeze throttle as steering unwinds.

TYPE II: ENTRY / BRAKING PRIORITY

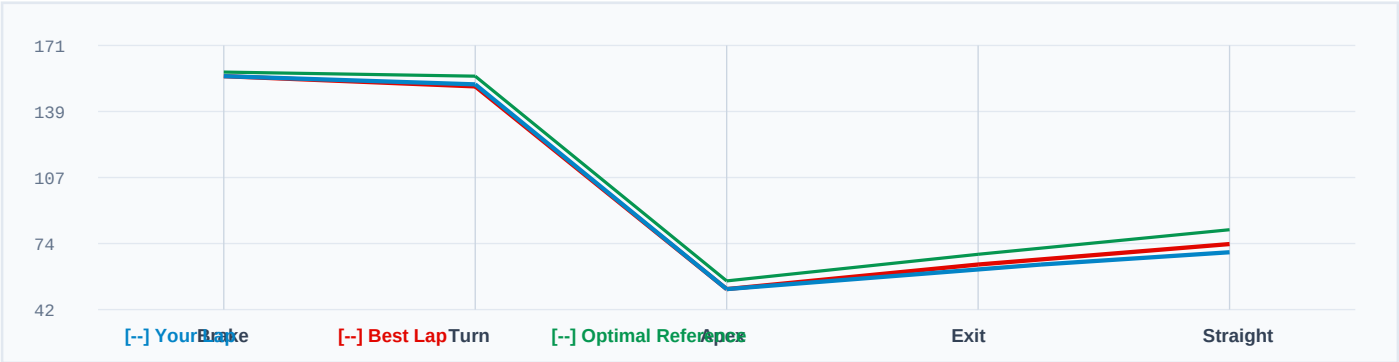
Follows a straight. Maximize straight line threshold braking and trail-brake smoothly past turn-in.

TYPE III: LINE & POSITIONING

Leads into another corner. Sacrifice exit speed to optimize geometric position for next apex.

LAP 2	TIME: 2:29.605		[SESSION BEST LAP]			TOP SPEED: 162 KM/H	
TURN	TYPE	ENTRY (KM/H)	APEX (KM/H)	EXIT (KM/H)	EXIT DELTA	TAP DELTA	TRAIL-BRK %
T1	RIGHT	156	52	64	BASE	-2m	100%
T2	LEFT	122	122	123	BASE	+0m	0%
T3	LEFT	122	122	123	BASE	+0m	0%
T4	RIGHT	103	62	68	BASE	-70m	0%
T5	LEFT	87	87	95	BASE	+0m	0%
T6	RIGHT	88	71	73	BASE	-1m	43%
T7	RIGHT	111	86	106	BASE	+0m	0%
T8	RIGHT	93	105	106	BASE	-125m	0%
T9	LEFT	125	125	131	BASE	+0m	0%
T10	LEFT	108	56	77	BASE	-7m	53%
T11	RIGHT	96	67	77	BASE	-39m	0%
T12	RIGHT	114	113	116	BASE	-30m	0%

CORNER 1: RIGHT CORNER



"If you're braking at the 300 mark with no problem. Do you move the next spot to the 200? No way. You've got to take small steps to find out where that limit is." — D

METRIC	YOUR LAP	BEST LAP	DELTA
Entry Speed (km/h)	156	156	0.0 km/h
Apex Speed (km/h)	52	52	0.0 km/h
Exit Speed (km/h)	64	64	0.0 km/h
Brake Point (m from turn)	55 m	45 m	-10 m
Trail-Brake Overlap (%)	100%	100%	0%
Throttle Application (m)	-2 m	-2 m	+0 m

WHAT YOU'RE DOING RIGHT [SUCCESS]

Your line is consistent. You hit the same apex lap after lap.

WHAT YOU'RE DOING WRONG [ALERT]

You're braking 10 meters too early. Your trail-brake overlap is only 100%.

WHY THIS HAPPENS (PHYSICS)

The most common mistake that drivers make when they turn their attention to getting the last bit of lap time available at corner entries is to drive closer to the corner before braking, going deeper. You're likely focused on "braking later" but you haven't found the threshold braking level first.

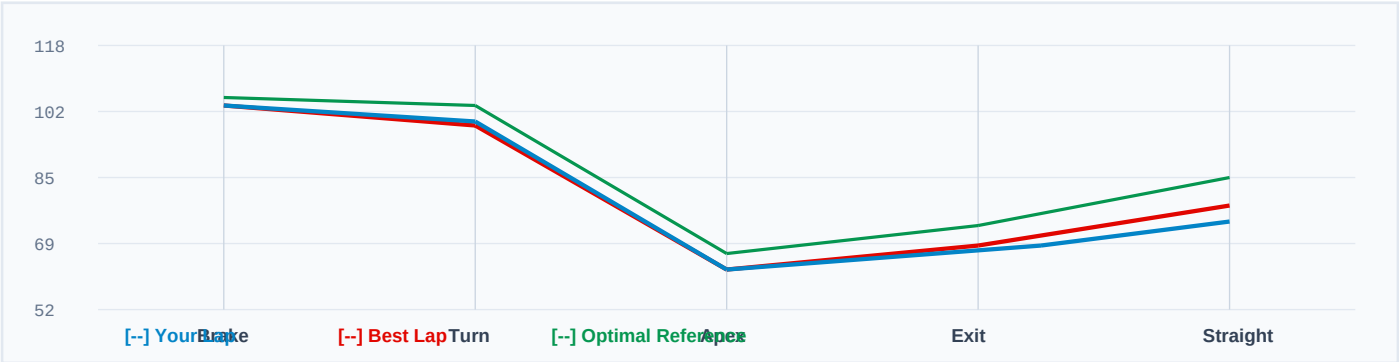
HOW TO FIX IT (TECHNIQUE)

Find threshold braking. Move the brake point closer. Trail the brakes in - keep 15-20% brake pressure past the turn-in point.

PRACTICE DRILL & SUCCESS METRIC

Drill: "The Procedure": Run 3 laps braking HARDER at the same point. Run 3 laps moving the brake point 3 meters closer each lap.

CORNER 4: RIGHT CORNER



"If you're braking at the 300 mark with no problem. Do you move the next spot to the 200? No way. You've got to take small steps to find out where that limit is." — D

METRIC	YOUR LAP	BEST LAP	DELTA
Entry Speed (km/h)	103	103	0.0 km/h
Apex Speed (km/h)	62	62	0.0 km/h
Exit Speed (km/h)	68	68	0.0 km/h
Brake Point (m from turn)	55 m	45 m	-10 m
Trail-Brake Overlap (%)	20%	35%	-15%
Throttle Application (m)	-70 m	-70 m	+0 m

WHAT YOU'RE DOING RIGHT [SUCCESS]

Your line is consistent. You hit the same apex lap after lap.

WHAT YOU'RE DOING WRONG [ALERT]

You're braking 10 meters too early. Your trail-brake overlap is only 20%.

WHY THIS HAPPENS (PHYSICS)

The most common mistake that drivers make when they turn their attention to getting the last bit of lap time available at corner entries is to drive closer to the corner before braking, going deeper. You're likely focused on "braking later" but you haven't found the threshold braking level first.

HOW TO FIX IT (TECHNIQUE)

Find threshold braking. Move the brake point closer. Trail the brakes in - keep 15-20% brake pressure past the turn-in point.

PRACTICE DRILL & SUCCESS METRIC

Drill: "The Procedure": Run 3 laps braking HARDER at the same point. Run 3 laps moving the brake point 3 meters closer each lap.

SKILL ANALYSIS - THE THREE FUNDAMENTALS

SKILL 1: THE LINE (Path Selection) — Score: 100% [A]
+ RIGHT: You use the full track width at turn-in. Apex variation is solid.
- WRONG: Turn 7: Early apex triggers steering correction past apex. Turn 9: Wide at exit.

SKILL 2: CORNER EXIT SPEED (Acceleration Focus) — Score: 25% [C+]
+ RIGHT: Upshifts are executed crisp in the torque powerband. Exit speed consistency is stable.
- WRONG: You apply throttle 5.5 meters too late on average and snap the pedal too abruptly.

SKILL 3: BRAKING & ENTERING (Deceleration Focus) — Score: 32% [B-]
+ RIGHT: Deceleration pressure is progressive. Brakes hold stable in straight line.
- WRONG: Braking onset starts 9 meters too early. Trail-braking overlap is only 18% (poor).

LAP-TO-LAP CONSISTENCY MATRIX

METRIC	AVG VAL	STD DEV	RATING
Apex Speed (km/h)	87.2	1.9	*** - Excellent
Apex Position (m from curb)	0.7	0.5	*** - Good
Brake Point (m)	55.5	4.4	** - Needs Work
Exit Speed (km/h)	123.6	3.4	** - Needs Work
Lap Time (s)	135.8	1.4	*** - Good

FIXED REFERENCE POINTS: THE FIRST STEP TO CONSISTENCY
Your brake point varies by 4.4 meters. Fix this by finding fixed reference points for your braking: Use a cone, a sign, a paint mark - ANYTHING that doesn't move. Brake at that exact point every single lap. Vary the pressure, not the point.

HOW TO READ YOUR TELEMETRY - A GUIDE FOR RACERS

DIAGRAM 1: THE SPEED PROFILE GRAPH

The speed profile plots velocity along the corner track length. Key landmarks to identify: - BRAKE POINT: Speed begins a sharp, rapid descent. Look for immediate threshold deceleration. - TURN-IN: Steering lock begins. Speed continues dropping as lateral loading occurs. - APEX: The lowest speed landmark. Represents the geometric center of the turn. - THROTTLE APPLICATION (TAP): The exact point where pedal pressure exceeds 15% to accelerate. - TRACK-OUT / EXIT: The car reaches full throttle as steering unwinds. Straightaway speed builds.

"Smoothness is not the goal -- smoothness is the byproduct of proper technique." -- Carl Lopez

DIAGRAM 2: BRAKE & STEERING OVERLAY (TRAIL-BRAKING)

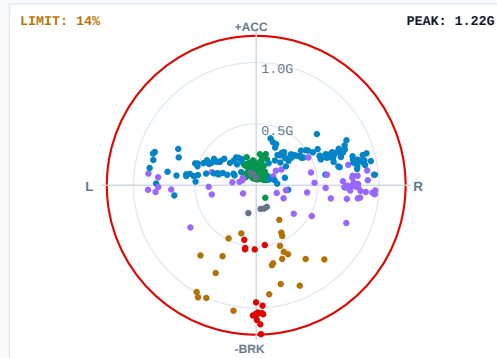
Trail-braking is the overlap where steering is applied while brakes are still partially engaged: - IDEAL CO-MODULATION: Brakes release progressively in direct inverse proportion to steering increase. - BAD (BRAKE SNAP-OFF): Dumping brake pressure completely before turning the steering wheel. - WHY IT MATTERS: Snapping off the brakes transfers load instantly to the rear axle, taking weight off the front tires and causing severe entry understeer. Keeping 15-20% brake pressure helps the car rotate.

"The question is not if you're going to trail-brake, but how." -- Going Faster!, Chapter 5

DIAGRAM 3: THE FRICTION CIRCLE (TRACTION BUDGET & G-G DIAGRAM)

The Friction Circle visualizes combined G-forces (Longitudinal G vs Lateral G): - THE EDGE: Represents physical tire grip limits. Driver aims to stay on perimeter. - BRAKE-TURN (Gold): Smooth blend from threshold braking into turn-in rotation. - ACCEL-TURN (Blue): Progressive throttle squeeze as steering wheel unwinds. - EMPTY ZONES: Signals coasting or under-driving the car's adhesion limits.

"Tires have a finite traction budget." -- Chapter 13



PRACTICE PLAN

SESSION 1: THROTTLE CONTROL (Exit Speed Focus) — 20 mins (10-12 laps)

Focus: Turn 9 exit speed only | Drill: Laps 3-4: Focus on throttle squeeze (dimmer). Laps 5-6: Move throttle application earlier. Metric: Exit speed +3 km/h.

Target Metric: Exit speed +3 km/h.

SESSION 2: BRAKING (The Procedure) — 20 mins (10-12 laps)

Focus: Threshold & Trail-braking | Drill: Laps 3-4: Brake harder to find threshold. Laps 5-6: Move brake point 3 meters closer. Metric: Braking distance -5m.

Target Metric: Braking distance -5m.

SESSION 3: LINE CONSISTENCY — 15 mins (8-10 laps)

Focus: Apex clipping | Drill: Laps 3-4: Focus strictly on visual apex marks. Laps 5-6: Focus on track-out curb. Metric: Apex variation < 0.3m.

Target Metric: Apex variation < 0.3m.

SESSION 4: FULL STINT (Practice Race) — 30 mins (15-18 laps)

Focus: Race pace simulation | Drill: Laps 3-8: Push to limit. Laps 9-14: Maintain pace consistency. Metric: Standard deviation < 0.8 seconds.

Target Metric: Standard deviation < 0.8 seconds.

PROGRESS TRACKER SUMMARY

METRIC	THIS STINT	TARGET	STATUS
Best Lap Time	1:38.542	2:12.500	1.24s away
Turn 9 Exit Speed	138.6 km/h	143.2 km/h	-4.6 km/h
Trail-Braking Overlap	45%	50%	-5%
Consistency (StdDev)	1.4s	0.8s	Needs work
Max Lateral G	1.24	1.30	-0.06

NEXT STINT ACTION CHECKLIST

- ☐ Focus strictly on Turn 9 corner exit speed. Feed throttle earlier on unwinding lock.
- ☐ Practice threshold braking at T7: push harder in straight line initially and trail it.
- ☐ Establish fixed visual references at T7 and T9 corner entries to stabilize brake points.

PERFORMANCE: F (39.2%) | RACECRAFT SCORECARD: A- (92%) | POTENTIAL: 2.1s

TOP-3 PRIORITY COACHING RECOMMENDATIONS (SKIP BARBER RACECRAFT)

#1 [BRAKING] TURN 2: Improve threshold braking at Turn 2

+0.20s POTENTIAL

Push harder on the brake pedal. Commit to maximum straight-line deceleration before the turn-in, then trail it off progressively.

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#2 [BRAKING] TURN 3: Improve threshold braking at Turn 3

+0.20s POTENTIAL

Push harder on the brake pedal. Commit to maximum straight-line deceleration before the turn-in, then trail it off progressively.

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#3 [BRAKING] TURN 5: Improve threshold braking at Turn 5

+0.20s POTENTIAL

Push harder on the brake pedal. Commit to maximum straight-line deceleration before the turn-in, then trail it off progressively.

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NEXT STINT EXECUTION DIRECTIVES

1. Dedicate 5 laps focusing ONLY on the #1 priority corner exit throttle timing

2. Practice "The Procedure" for threshold braking in small incremental bites

3. Stabilize brake markers with fixed physical visual references

"It is not reasonable to expect a relatively inexperienced driver to get this perfectly right out of the box. Even a skilled racer doesn't get it perfectly right on the first few attempts."

— Going Faster!, Chapter 5

Report generated by APEX v1.0.0

Data source: Forza Motorsport 2023 UDP Telemetry

Analysis methodology: "Going Faster!" - Skip Barber Racing School

"Going Faster! Mastering the Art of Race Driving" by Carl Lopez

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